

Instructions

1. Identify all entities
2. Identify primary keys
3. Identify foreign keys
4. Identify relationships
5. Show cardinality

1. Entities

- **Entities are the main things/objects the system needs to track. From the case study, we have 7 core entities:**
 - Customers
 - Products
 - Orders
 - Order position
 - Payments
 - Deliveries
 - Suppliers

2. Primary Keys

- **unique identifier for each record in a table. consisting of the entity name and usually ends with ID.**
 - Customers - Customer_ID
 - Products – Product_ID
 - Orders - Order_ID
 - Order Line - OrderLine_ID
 - Payments - Payment_ID
 - Deliveries – Delivery_ID
 - Suppliers - Supplier_ID

3. Foreign keys and

- Field in one table that links to the Primary Key in another table.
- Customer → Address:
One customer, many addresses — addresses cannot exist without a customer
- Customer → Order:
Customers may place unlimited orders; orders are owned by exactly one customer
- Order → Payment:
Enforced 1:1 as required — every order has exactly one payment record
- Order → Delivery:
Enforced 1:1 every order has exactly one delivery record
- Order → Order Line:
One order contains many products — resolved via junction table to avoid M: N
- Address → Delivery:
Deliveries reference one registered address; an address may be reused across multiple orders